3-4-26 (WEEK 2)

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Suppose $R, S$: rings with a ring homomorphism $\rho: R \to S$

**DEFINITION**

In particular, $J$: prime $\Rightarrow \varphi^{-1}(J)$: prime. $\Rightarrow$ $\begin{array}{ccc} \operatorname{Spec} S & \xrightarrow{\operatorname{Spec}(\varphi)} & \operatorname{Spec} R \\ J & \longmapsto & \varphi^{-1}(J) \end{array}$

Suppose $J \trianglelefteq S$ over $\varphi^{-1}(J) \trianglelefteq R$. We say $J$ lies over $I$ if $I = \varphi^{-1}(J)$.

**PROPOSITION**

Let $A \subseteq B$ be an integral extension of rings.
Let $p \trianglelefteq A$ be a prime ideal, then:
① $pB \neq B$
② $\exists$ a prime ideal $Q \trianglelefteq B$ s.t. $Q \cap A = p$

**REMARK (EXERCISE)**

$\operatorname{Spec}(\varphi)$ is a cont. map w.r.t. Zariski topology

As $A \subseteq B$, $\exists$ natural inclusion $i: A \to B$. So $Q \trianglelefteq B$, $i^{-1}(Q) = Q \cap A$
Then, Prop $\Leftrightarrow$ $\operatorname{Spec} B \leftrightarrow \operatorname{Spec} A$ surj when $A \subseteq B$: int ext

**PROOF OF PROPOSITION**

② $\Rightarrow$ ①, so it suffices to show ②
Let $S = A \setminus p \subseteq B$, then let $S^{-1}A = A_p$, $S^{-1}B = B_p$

$\varphi$: not necessarily inj, but we have:

$$\begin{tikzcd} A \arrow[r, "\varphi"] \arrow[d, hook] \arrow[dr, phantom, "\circlearrowleft"] & A_p \arrow[d, hook] \\ B \arrow[r, "\varphi"] & B_p \end{tikzcd}$$

As $A_p \hookrightarrow B_p$: inclusion, thus $B_p$: integral over $A_p$.

Now, let $\tilde{Q}$ be a max ideal of $B_p$
Thus, $\tilde{Q} \cap A$: max ideal of $A_p$ (by prop last time)
However, $A_p$ has only one max ideal $pA_p$
$\therefore$ $\tilde{Q} \cap A_p = pA_p$
$\Rightarrow \varphi^{-1}(\tilde{Q} \cap A_p) = \varphi^{-1}(pA_p) \Rightarrow \varphi^{-1}(\tilde{Q}) \cap A = pA$
$\therefore$ $\tilde{Q}$: prime $\Rightarrow \varphi^{-1}(\tilde{Q})$: prime in $B$. $\square$

**CONCLUSION**

$K/\mathbb{Q}$: number field $\Rightarrow \mathcal{O}_K$: ring

**GOAL**

$\mathcal{O}_K$: f.g. $\mathbb{Z}$-mod

# FINITENESS PROPERTIES OF $\mathcal{O}_K$, PRIMES IN $\mathcal{O}_K$

**LEMMA**

Suppose $K/\mathbb{Q}$: number field. Then, $\forall \alpha \in K \setminus \{0\}$, $\exists d \in \mathbb{Z}_{>0}$ s.t. $d\alpha \in \mathcal{O}_K$

Proof

Let $f_{\alpha,\mathbb{Q}} := \sum_{i=0}^{n} c_i x^i$ be the min poly of $\alpha$ over $\mathbb{Q}$

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As $c_i \in \mathbb{Q}$ $\forall i$, $\exists d \in \mathbb{Z}_{>0}$, s.t. $dc_i \in \mathbb{Z}$

Then, $d^n f_{\alpha,\mathbb{Q}}(x) = \sum_{i=0}^{n} \underbrace{c_i d^{n-i}}_{\in \mathbb{Z}} (xd)^i$

$\therefore$ $\alpha d$ is a root of $\sum_{i=0}^{n} c_i d^{n-i} T^i \in \mathbb{Z}[T]$, $c_n = 1$

$\therefore$ $\alpha d \in \mathcal{O}_K$ $\square$

In particular, we can choose a basis $\{e_i\}$ of $K/\mathbb{Q}$ with $e_i \in \mathcal{O}_K$

## DEFINITION (TRACE AND NORM)

Let $k$: field, $K/k$: finite ext

$\forall a \in K^\times$: Consider $\begin{aligned} m_a : K &\longrightarrow K \\ \alpha &\longmapsto a\alpha \end{aligned}$ (a $k$-linear map)

Choose a basis $\{e_i\}_{i=1}^{n}$ of $K/k$. Let $M_a$ be the matrix representation w.r.t. $\{e_i\}$

Then, we define: $\mathrm{Tr}_{K/k}(a) = \mathrm{Tr}(M_a)$

$\mathrm{Nr}_{K/k}(a) = \det(M_a)$

Note, $\mathrm{Tr}_{K/k}(a)$ and $\mathrm{Nr}_{K/k}(a)$ are indep of choice of $\{e_i\}$

## MAIN THEOREM

$\mathcal{O}_K$: f.g. $\mathbb{Z}$-mod

Note, $K$: field, $\mathcal{O}_K$-torsion free $\mathbb{Z}$-mod

By Structure Thm of modules over PID, $\mathcal{O}_K$: f.g. free $\mathbb{Z}$-mod, i.e. $\mathcal{O}_K \cong \mathbb{Z}^n$ as $\mathbb{Z}$-mod, $n = [K:\mathbb{Q}]$

## LEMMA

Fix $a \in K^\times$, let $f_{a,k} = \sum_{i=0}^{n} c_i x^i$ be the min poly of $a$ over $k$. Then, $\mathrm{Tr}_{k(a)/k}(a) = -c_{n-1}$

Proof

Notice $\{1, a, \dots, a^{n-1}\}$ is a basis of $k(a)/k$

Then, $M_a = \begin{pmatrix} 0 & \cdots & 0 & -c_0 \\ 1 & 0 & & \vdots \\ & 1 & \ddots & \\ 0 & & 1 & -c_{n-1} \end{pmatrix} \Rightarrow \mathrm{Tr}_{k(a)/k}(a) = -c_{n-1}$

## LEMMA

$\forall a \in K^\times$, $\mathrm{Tr}_{K/k}(a) = [K:k(a)]\,\mathrm{Tr}_{k(a)/k}(a)$

Proof

Let $\{e_i\}_{i=1}^{m}$ be a basis of $K/k(a)$

Then, $\{e_i a^j\}_{1 \le i \le m, 1 \le j \le n-1}$ is a basis of $K/k$

Notice, $m_a : e_i a^j \longmapsto e_i a^{j+1}$

$\therefore$ $M_a = \begin{pmatrix} \ulcorner & & \\ & \ddots & \\ & & \ulcorner \end{pmatrix}$ ($e_1$, $e_m$)

Here, each block $V_i$ is the subspace of $K$ generated by $\{e_i a^j\}_{j=1}^{n-1}$, where $m_a : V_i \to V_i$

$\therefore$ $\mathrm{Tr}(M_a|_{V_i}) = -c_{n-1} \Rightarrow \mathrm{Tr}_{K/k}(a) = [K:k(a)]\,\mathrm{Tr}_{k(a)/k}(a)$ $\square$

## LEMMA

If $K/k$ is inseparable, then $\mathrm{Tr}_{K/k} = 0$

Proof

Fix $a \in K^\times$. Then, we have int ext $k \subseteq k(a) \subseteq K$

So, $K/k$: insep $\Rightarrow$ $K/k(a)$: insep or $k(a)/k$: insep

① Suppose $K/k(a)$: insep. Then, $\exists \alpha \in K^\times$, s.t. $f_{\alpha,k(a)}(x)$: not sep (multiple root)

$\therefore$ $f'_{\alpha,k(a)}(x) = 0$, i.e. $f_{\alpha,k(a)} = \sum_{i=0}^{n} c_i x^{p_i}$

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$\Rightarrow p \mid [k(a,\alpha):k(\alpha)] \Rightarrow p \mid [K:k(\alpha)]$

By lemma, $\mathrm{Tr}_{K/k}(a)=0$

② Suppose $k(a)/k$: insep

Then, $f_{a,k}(x)=\sum_{i=0}^{n} c_i x^{p^i}$

By lemma, $\mathrm{Tr}_{k(a)/k}(a)=\text{coef of } x^{p^i-1}=0$

By lemma, $\mathrm{Tr}_{K/k}(a)=0$ □

## LEMMA

Let $K/k$: finite sep ext, $k\subseteq K\subseteq L$, $L/k$: finite Galois ext

Then, $\forall \alpha\in K^\times$,

$$\mathrm{Tr}_{K/k}(a)=\sum_{\substack{\sigma:\text{ runs over}\\ k\text{-embedding } K\hookrightarrow L}} \sigma(a)$$

$\left(\begin{array}{l}\forall \tau\in \mathrm{Gal}(L/k),\ \tau\cdot \mathrm{Tr}_{K/k}(\alpha)=\mathrm{Tr}_{K/k}(\alpha)\\ \text{By Gal Thry, } \mathrm{Tr}_{K/k}(\alpha)\in k\end{array}\right)$

<u>Proof of Special Case</u>

By primitive element thm, $\exists\theta\in K$, s.t. $K=k(\theta)\cong k[x]/f_{\theta,k}(x)$

Assume $L$ is the splitting field of $f_{\theta,k}(x)$.

Notice, $\underset{\sigma_i}{\{k\text{-embedding } K\hookrightarrow L\}} \overset{1-1}{\longleftrightarrow} \underset{\sigma_i(\theta)}{\{\text{roots of } f_{\theta,k}(x) \text{ in } L\}}$

By lemma, $\mathrm{Tr}_{K/k}(\theta)=-c_{n-1}=\sum_{i=0}^{n-1}\sigma_i(\theta)$

Proof of General Case: Galois Theory + $\mathrm{Tr}_{K/k}(a)=[K:k(a)]\,\mathrm{Tr}_{k(a)/k}(a)$

## COROLLARY

Given finite sep exts $k\subseteq K\subseteq L$, we have $\mathrm{Tr}_{L/k}=\mathrm{Tr}_{K/k}\circ\mathrm{Tr}_{L/K}$

## DEFINITION

Say $K/k$: finite, sep

We define sym bilinear form $\langle_,_\rangle: K\times K\longrightarrow k$

$$(x,y)\longmapsto \mathrm{Tr}_{K/k}(xy)$$

## THEOREM

$\{e_i\}$: basis, $M$: matrix rep of $\langle_,_\rangle$ w.r.t. $\{e_i\}$, $a_{ij}=\langle e_i,e_j\rangle$

Then, $\langle v,w\rangle=w^tMv$, $\det(M)\neq 0$ (well-def in $k^\times/k^{\times 2}$)

If $K/k$: finite sep, then $\langle_,_\rangle$: <u>nondegenerate</u>

<u>Proof</u>

$K/k$: finite sep $\Rightarrow$ By primitive element thm, $\exists\theta\in K^\times$, s.t. $K=k(\theta)$

Choose a basis for $K/k$: $\{1,\theta,\dots,\theta^{n-1}\}$, $n=[K:k]$

Let $M=(a_{ij})$ be the matrix rep of $\langle_,_\rangle$ w.r.t. $\{1,\theta,\dots,\theta^{n-1}\}$

Then, $a_{ij}=\mathrm{Tr}_{K/k}(\theta^{i+j})=\sum_{\sigma:k\hookrightarrow L}\sigma(\theta^{i+j})$, $L$: splitting field of $f_{\theta,k}$

Let $\{\sigma_0,\dots,\sigma_{n-1}\}$ be the set of $k$-embedding $K\hookrightarrow L$

Set $T=(t_{ij})$, $t_{ij}=\sigma_j(\theta^i)_{0\le i,j\le n-1}$

Then, $a_{ij}=(TT^t)_{ij}=\sum_r t_{ir}t_{jr}=\sum_r\sigma_r(\theta^{i+j})$

$\Rightarrow M=TT^t\Rightarrow \det(M)=\det(T)^2$

$\therefore \det(M) = \left[\prod_{i \neq j} (\sigma_i(\theta) - \sigma_j(\theta)) \cdot (-1)^{\frac{n(n-1)}{2}}\right]^2 \neq 0$ □

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## PROOF OF MAIN THEOREM

Idea: By original lemma, we can choose a basis $\{e_i\}$ of $K/\mathbb{Q}$ s.t. $e_i \in \mathcal{O}_K$

$\left(\begin{array}{l} \forall \alpha \in \mathcal{O}_K,\ \mathbb{Q} \subseteq K \subseteq L,\ L/\mathbb{Q}: \text{Gal closure of } K \Rightarrow \sigma(\alpha) \in \mathcal{O}_L\ \forall \sigma \in \mathrm{Gal}(L/\mathbb{Q}) \\ f_{\alpha,K} = \sum c_i x^i \in \mathbb{Z}[x],\ \text{monic}.\ \sigma(\alpha) \text{ is a root of } f_{\alpha,K} \Rightarrow \sigma(\alpha) \in \mathcal{O}_L \end{array}\right)$

Fix $x \in \mathcal{O}_K$, then $x = \sum_{i=1}^{n} c_i e_i$, $c_i \in \mathbb{Q}$

To find $d \in \mathbb{Z}_{>0}$ indep of $x$ s.t. $c_i d \in \mathbb{Z}\ \forall i$, i.e. $x \overset{?}{=} \sum (c_i d)\left(\frac{e_i}{d}\right)$

Then, $\langle x, e_i \rangle = \mathrm{Tr}_{K/\mathbb{Q}}(\underbrace{xe_i}_{\in \mathcal{O}_K})$

$$= \underbrace{\sum_{\substack{\sigma: K \hookrightarrow L \\ K:\text{emb}}} \underbrace{\sigma(xe_i)}_{\in \mathcal{O}_L}}_{\in \mathcal{O}_L \cap \mathbb{Q} = \mathbb{Z}}$$

In other words, $\langle x, e_i \rangle = (c_1, \dots, c_n) M \begin{pmatrix} 0 \\ \vdots \\ 1 \\ \vdots \\ 0 \end{pmatrix} = (0, \dots, \overset{i\text{th}}{1}, \dots, 0) M \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} \Rightarrow M \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} \in \mathbb{Z}^n$

$\therefore \underbrace{M^t M}_{(a_{ij}) = \langle e_i, e_j \rangle \text{ must thus be in } \mathbb{Z}} \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} = \det(M) \begin{pmatrix} c_1 \\ \vdots \\ c_n \end{pmatrix} \in \mathbb{Z}^n$

So, let $d = |\det(M)|$, we will have $d c_i \in \mathbb{Z}$ indep of $x \in \mathcal{O}_K$.

$\therefore \bigoplus \mathbb{Z} e_i \subseteq \mathcal{O}_K \subseteq \bigoplus \mathbb{Z}\left(\frac{e_i}{d}\right)$

By structure thm of mod over PID, $\mathcal{O}_K$: free $\mathbb{Z}$-mod with rank $n = [K:\mathbb{Q}]$ □